

The Flower's Wedding

II PUC · Ch 1 · Sexual Reproduction in Flowering Plants

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



The Invitation

A hibiscus flower sent Bindu a card: "You are invited to our wedding — **sexual reproduction**!" Prof. BAIO explained: "The groom's side is the **stamen** — filament and anther. The bride's side is the **pistil** — stigma, style and ovary."

💡 **REMEMBER**

Androecium = stamens (male) · Gynoecium = pistils (female).



The Pollen Factory

Inside each anther, four **microsporangia** were busy. Their walls had four layers: epidermis, **endothecium**, middle layers and the nourishing **tapetum**. Microspore mother cells divided by meiosis — **microsporogenesis** — making tetrads of microspores.

💡 **REMEMBER**

Tapetum cells have dense cytoplasm and often more than one nucleus; they nourish the pollen.



The Toughest Coat

Each pollen grain wore an **exine** made of **sporopollenin**, the most resistant organic material known — strong acids and heat cannot destroy it! Where the exine was missing were **germ pores**. Inside were a big vegetative cell and a small **generative cell**.

💡 **REMEMBER**

Over 60% of angiosperms shed pollen at the 2-celled stage.



The Bride’s Palace

Inside the ovary sat the **ovule** with its integuments, a tiny opening called the **micropyle** and the **nucellus**. A **megaspore mother cell** divided by meiosis; one functional megaspore grew into the **embryo sac** — 8 nuclei in 7 cells.

💡 **REMEMBER**

Egg apparatus (egg + 2 synergids) at the micropylar end, 3 antipodals at the chalazal end, 2 polar nuclei in the central cell.



Travel Plans

Pollen had to travel to the stigma — by wind, water or insects. **Autogamy** meant the same flower, **geitonogamy** another flower of the same plant, and **xenogamy** a flower of another plant. Viola even had closed **cleistogamous** flowers that never open.

💡 **REMEMBER**

Geitonogamy works like cross-pollination but is genetically similar to autogamy.



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No Marrying Cousins

Flowers did not like inbreeding, so they used **outbreeding devices**: pollen and stigma ready at different times, anther and stigma at different heights, **self-incompatibility**, and separate male and female flowers.

💡 REMEMBER

Castor and maize are monoecious; papaya is dioecious.



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The Double Wedding

The **pollen tube** grew down the style and entered a synergid through the **filiform apparatus**. It released two male gametes. One fused with the egg — **syngamy** — making the zygote. The other fused with the two polar nuclei — **triple fusion** — making the **primary endosperm nucleus**. Two fusions: **double fertilisation**!

💡 **REMEMBER**

Zygote (2n) → embryo · PEN (3n) → endosperm.



Happily Ever After

The ovule became a seed and the ovary became a fruit. Coconut water is free-nuclear **endosperm**. Some plants made seeds without fertilisation — **apomixis**; Citrus made many embryos in one seed — **polyembryony**; banana made fruits without fertilisation — **parthenocarpy**.

💡 **REMEMBER**

Albuminous seeds keep endosperm (wheat, maize, castor); non-albuminous seeds use it up (pea, groundnut).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

sexual reproduction p.1 · stamen p.1 · pistil p.1 · microsporangia p.2 · endothecium p.2 · tapetum p.2 · microsporogenesis p.2 · exine p.3 · sporopollenin p.3 · germ pores p.3 · generative cell p.3 · ovule p.4 · micropyle p.4 · nucellus p.4 · megaspore mother cell p.4 · embryo sac p.4 · Autogamy p.5 · geitonogamy p.5 · xenogamy p.5 · cleistogamous p.5 · outbreeding devices p.6 · self-incompatibility p.6 · pollen tube p.7 · filiform apparatus p.7 · syngamy p.7 · triple fusion p.7 · primary endosperm nucleus p.7 · double fertilisation p.7 · endosperm p.8 · apomixis p.8 · polyembryony p.8 · parthenocarpy p.8



QUICK QUIZ

Do you remember?

1. The most resistant organic material is:

- Cellulose
- Sporopollenin
- Lignin
- Pectin

2. A typical embryo sac has:

- 8 nuclei, 7 cells
- 7 nuclei, 8 cells
- 8 nuclei, 8 cells
- 4 nuclei, 4 cells

3. Triple fusion forms the:

- Zygote
- Primary endosperm nucleus
- Embryo
- Seed coat

4. The filiform apparatus is found in:

- Antipodals
- Synergids
- Egg
- Polar nuclei

5. Fruit formation without fertilisation is:

- Apomixis
- Polyembryony
- Parthenocarpy
- Geitonogamy

Answers: 1-Sporopollenin · 2-8 nuclei, 7 cells · 3-Primary endosperm nucleus · 4-Synergids · 5-Parthenocarpy